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LETTER TO THE EDITOR

High-energy heavy-ion irradiation of $\text{Fe}_{85}\text{B}_{15}$ amorphous alloy: effect of an external applied stress

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Abstract. The growth phenomenon observed in amorphous metallic alloys irradiated with energetic heavy ions is now well established but yet poorly understood. The experiment reported here shows, for the first time, that an external applied stress induces a drastic amplification of the phenomenon.

Up to a few years ago, it was commonly admitted that the electronic energy loss of energetic ions could have no effect on radiation damage creation in materials presenting a metallic character (contrary to the case of insulators) because the density of free electrons was too high. However, it was recently demonstrated that electronic energy loss induces a recovery of the defects created by nuclear collisions in pure transition metals (Iwase *et al* 1987, Dunlop *et al* 1987). In amorphous metallic alloys, the effects of electronic energy loss are even more drastic: (i) defect production with a cross section two orders of magnitude higher than that of elastic displacements (Audouard *et al* 1987) and (ii) large anisotropic growth (Audouard *et al* 1988, Klaumünzer *et al* 1987) were observed during irradiation with high-energy heavy ions. Above an incubation fluence, during which damage is created in the alloy, the sample dimensions perpendicular to the ion beam increase while the sample dimension parallel to it shrinks, presumably without volume change (Klaumünzer *et al* 1986, Schumacher *et al* 1984). Similar growth effects have recently been reported in insulating amorphous oxides (Klaumünzer *et al* 1988).

Although the microscopic mechanism responsible for these effects is not yet understood, the anisotropic growth phenomenon could be related to an irradiation-induced local stress: as an incident ion passes through the sample, the surrounding atoms receive an energy pulse for a very short time. A general description of a plastic

deformation induced by a short localised energy pulse was given, several years ago, by Bullough and Gilman (1966). Although the relationship between the observed anisotropic deformation of irradiated amorphous metallic alloys and the induced local stress is not straightforward, we can assume that the growth is due to the relaxation of this local stress.

Klaumünzer and Schumacher (1983) postulate that the ion explosion spike produces a strong transient shear stress, leading to a plastic deformation of the sample when the ion irradiation has induced a sufficient amount of free volume to allow atomic mobility. The question arises, however, of the intensity of this local stress. One way to answer this question is to study the influence of an external applied stress on the growth rate of irradiated amorphous metallic alloys.

Two amorphous 21 μm thick $\text{Fe}_{85}\text{B}_{15}$ ribbons were irradiated in the same run at a temperature of 80 K with 2.8 GeV Xe ions delivered by the French Grand Accélérateur National d'Ions Lourds (GANIL) at Caen. In this case, the electronic stopping power is $S_e = 6$ keV per interatomic spacing. The total Xe fluence Φ was 1.7×10^{13} ions/ cm^2 , which corresponds to a total number of displacements per atom (DPA) created by elastic collisions of 2.5×10^{-4} . The induced radiation effects were studied by *in situ* electrical resistance measurements. One of the samples was submitted to an external uniaxial tensile stress $\sigma = 600$ MPa, parallel to the DC measurement current, and the other was unstressed. It has been verified, prior to irradiation, that during a few days at 80 K the electrical resistance of the stressed sample remained constant. During the experiments, we have also checked, at fluences of 3×10^{12} and 1.7×10^{13} ions/ cm^2 , the long term ($\sim 10^4$ s) stability of the electrical resistance out of irradiation. The observed resistance variations were less than $\sim 10^{-4}$ for both stressed and unstressed samples, i.e. much less than the amplitude of the growth effect. In other words, we have consequently verified that there is no creep at 80 K out of irradiation.

The variations in electrical resistance of the two samples, $\Delta R/R_0$, are plotted against the Xe fluence in figure 1. The result obtained on the unstressed sample reproduces well that already reported by Audouard *et al* (1988). In contrast, a huge effect of the external applied stress is observed on the second sample. The part of the

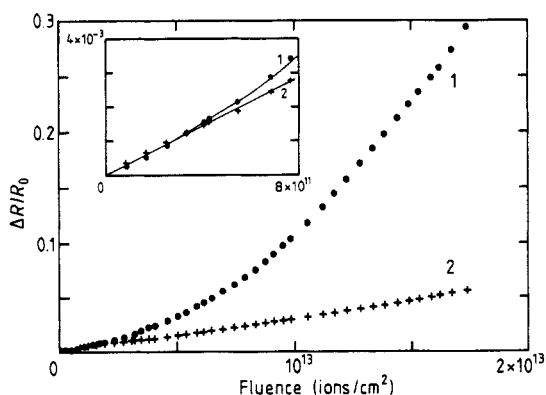


Figure 1. Relative resistance variation $\Delta R/R_0$ plotted against Xe fluence for $\text{Fe}_{85}\text{B}_{15}$ amorphous alloys irradiated at 80 K with 2.8 GeV Xe ions. Sample 1 is submitted to an external tensile stress $\sigma = 600$ MPa and sample 2 is unstressed. The inset presents the beginning of the irradiation curve.

curve where the increase of the electrical resistivity, due to defect creation, dominates the resistance variation (low irradiation fluence (Audouard *et al* 1988)) is not perceptibly modified by the applied stress (see the inset to the figure), while a drastic increase of $d(\Delta R/R_0)/d\Phi$ from $4 \times 10^{-15} \text{ cm}^2$ (for $\sigma=0$) to $34 \times 10^{-15} \text{ cm}^2$ (for $\sigma=600 \text{ MPa}$) is observed in the growth region (high irradiation fluence).

The growth phenomenon is the macroscopic manifestation of ion-beam-induced atomic transport. The fact that no significant resistance variation in the stressed sample was observed out of irradiation is an indication that the atomic transport is not due to the presence of defects alone but is the consequence of the conjunction of two conditions: (i) the presence of defects (incubation period) and (ii) a transient increase of their mobility around the ion path. The experiment described here shows clearly that the amplitude of the phenomenon is drastically amplified by the external applied stress. Let us introduce a number n_g defined as follows: one incident ion covering the distance of an average interatomic spacing induces the motion of n_g atoms involved in the growth phenomenon. This number n_g can be deduced from experimental data in the high-fluence growth region, assuming that, in this region, (i) the variation of R comes only from the variation in shape factor (i.e. variations in electrical resistivity have a negligible effect (Audouard *et al* 1988)) and (ii) the growth phenomenon occurs without volume change (Klaumünzer *et al* 1986, Schumacher *et al* 1984). Then

$$n_g = \frac{d(\Delta R)}{R_0 d\Phi} Na \quad (1)$$

where N is the atomic density and a the interatomic spacing of the irradiated alloy and where $d(\Delta R)/R_0 d\Phi$ is calculated in the steady-state growth regime (irradiation fluence higher than $10^{13} \text{ ions/cm}^2$). We find $n_g=8$ for the unstressed sample. This value is in good agreement with the values deduced from the results of Xe-irradiated $\text{Pd}_{80}\text{Si}_{20}$ (Klaumünzer *et al* 1986): n_g lies between 7 ($S_c=2.8 \text{ keV/interatomic spacing}$) and 19 ($S_c=13 \text{ keV/interatomic spacing}$). For the stressed sample, n_g is certainly much higher. To evaluate it, the amplitude of the induced strain has to be measured along and perpendicular to the stress direction: experiments are now in progress.

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